



PRODUCT NAME

Industrial-grade Debug Test 3 Formula

Description

A professionally developed industrial-grade cream formulated for debugging and testing purposes. This cost-effective, unscented product provides a medium viscosity suitable for various industrial applications while ensuring safety and quality.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	50-100 kg
Processing Time:	2-3 hours
Temperature:	70°C for mixing phases
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessels, heat source, stirrers
Certification:	ISO 9001, GMP compliant

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Cetyl Alcohol	Cetyl Alcohol	10.0%	Emulsifying Agent
3	Glycerin	Glycerin	5.0%	Humectant
4	Stearic Acid	Stearic Acid	8.0%	Thickener
5	Glyceryl Stearate	Glyceryl Stearate	5.0%	Emulsifying Agent
6	Mineral Oil	Paraffinum Liquidum	8.0%	Moisturizer
7	Triethanolamine	Triethanolamine	2.0%	pH Adjuster
8	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
9	Tocopheryl Acetate	Tocopheryl Acetate	1.0%	Antioxidant
Total			100%	

Manufacturing Process

Phase 1: Phase A

1. Combine Water, Glycerin, and Triethanolamine in a large mixing vessel.
2. Heat the mixture to 70°C while stirring continuously.

Phase 2: Phase B

1. In a separate container, combine Cetyl Alcohol, Stearic Acid, Glyceryl Stearate, and Mineral Oil.
2. Heat the Phase B ingredients to 70°C until fully melted.

Phase 3: Emulsification

1. Slowly add Phase B to Phase A while maintaining temperature and stirring vigorously to create an emulsion.
2. Continue stirring until the mixture is homogeneous.

Phase 4: Cooling and Finishing

1. Cool the emulsion to 40°C with continued mild stirring.
2. Add Phenoxyethanol and Tocopheryl Acetate.
3. Stir until fully combined and ensure the final product is smooth and consistent.

Required Equipment

Mixing vessels, heat source, stirrers

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001, GMP compliant

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount to desired surfaces or equipment as needed for debugging purposes. Ensure surfaces are clean before application.